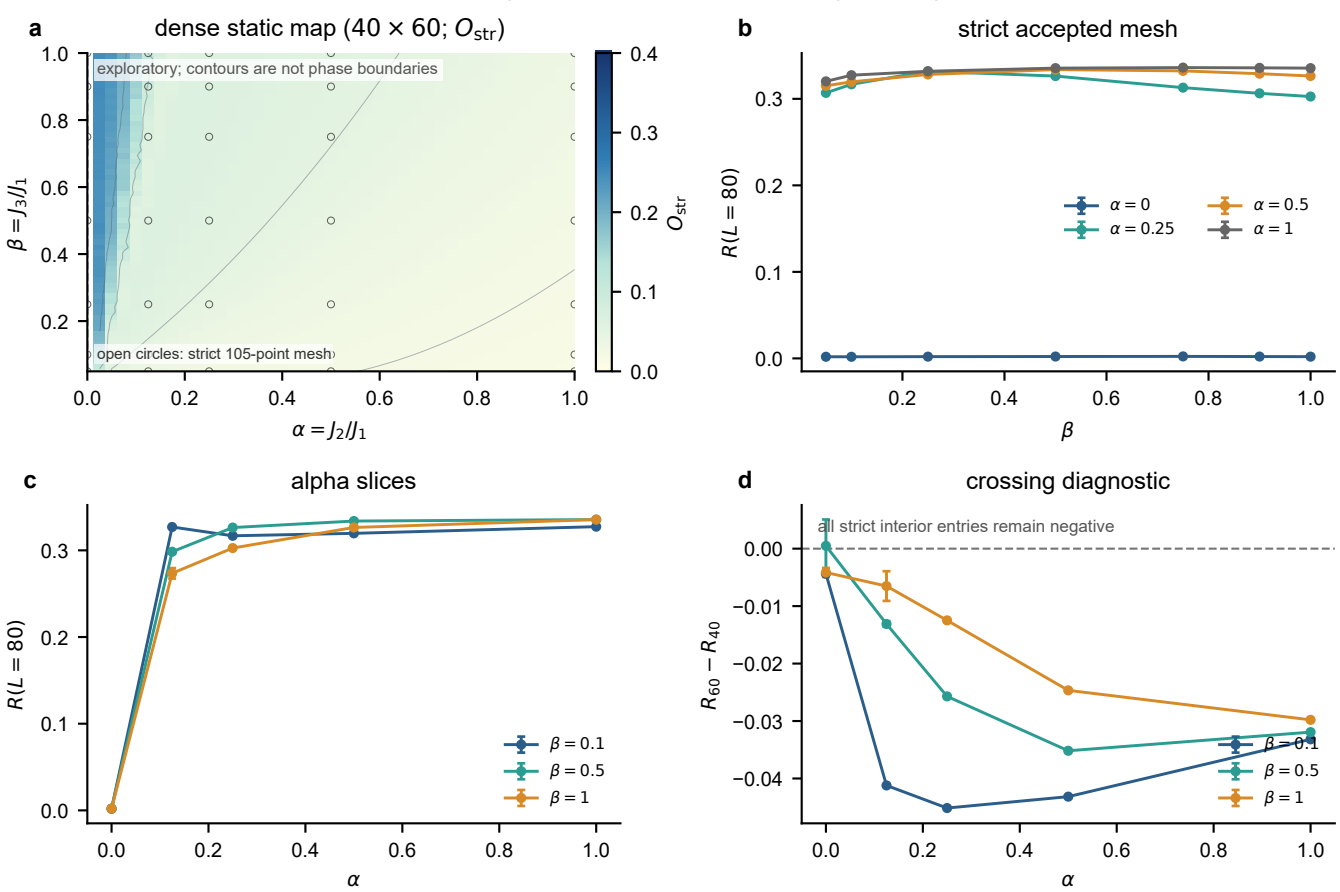

1×5 Heisenberg chain: refined positive-region diagnostics
a
dense static map (40 × 60; O_{str})
exploratory; contours are not phase boundaries
open circles: strict 105-point mesh
$\beta = J_3/J_1$
$\alpha = J_2/J_1$
O_{str}
b
strict accepted mesh
$R(L = 80)$
β
$\alpha = 0$
$\alpha = 0.25$
$\alpha = 0.5$
$\alpha = 1$
c
alpha slices
$R(L = 80)$
α
$\beta = 0.1$
$\beta = 0.5$
$\beta = 1$
d
crossing diagnostic
all strict interior entries remain negative
$R_{60} - R_{40}$
α
$\beta = 0.1$
$\beta = 0.5$
$\beta = 1$